Data Analyst Capstone Project
Technology Trends Analysis using Stack Overflow Survey

Meghna Sharma

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Skills Network

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OUTLINE

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EXECUTIVE SUMMARY

Summarize key findings here

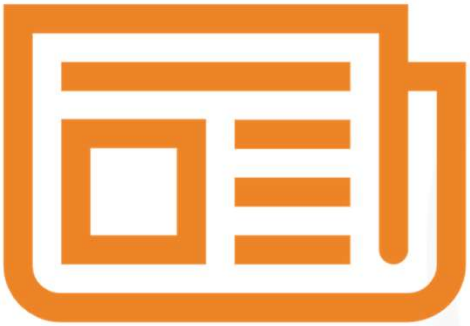
- Analysis of the **Stack Overflow Developer Survey dataset** to understand technology usage and trends.
- Key technologies currently used by developers
  - **JavaScript, SQL,** and **Python** are among the most widely used programming languages.
  - **PostgreSQL** and **MySQL** are popular database systems.
  - Cloud platforms like AWS and Azure dominate platform usage.
- Future technology trends show increasing interest in modern tools such as **Rust, TypeScript,** and cloud platforms.
- Developer demographics indicate most respondents are between **25–34 years old** with **Bachelor's or Master's degrees**.
- Dashboards built using **IBM Cognos Analytics** provide interactive insights into technology usage, trends, and demographics.

INTRODUCTION

Purpose of the report, Target Audience & Value of the analysis

- The purpose of this report is to analyze developer survey data to understand current technology usage and emerging technology trends.
- The analysis focuses on programming languages, databases, platforms, web frameworks, and developer demographics.
- The target audience includes students, developers, recruiters, and organizations interested in technology trends in the software industry.
- The value of this analysis includes:
  - Identifying widely used technologies in the industry.
  - Understanding future technology preferences among developers.

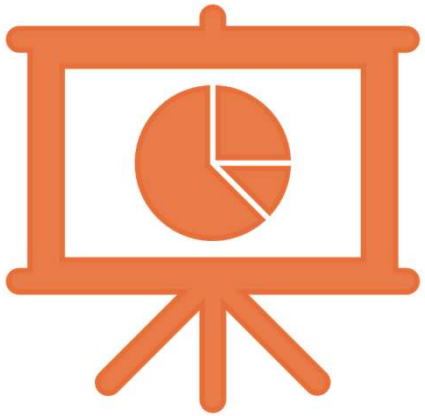
METHODOLOGY



Data sources, Data collection & Data wrangling steps

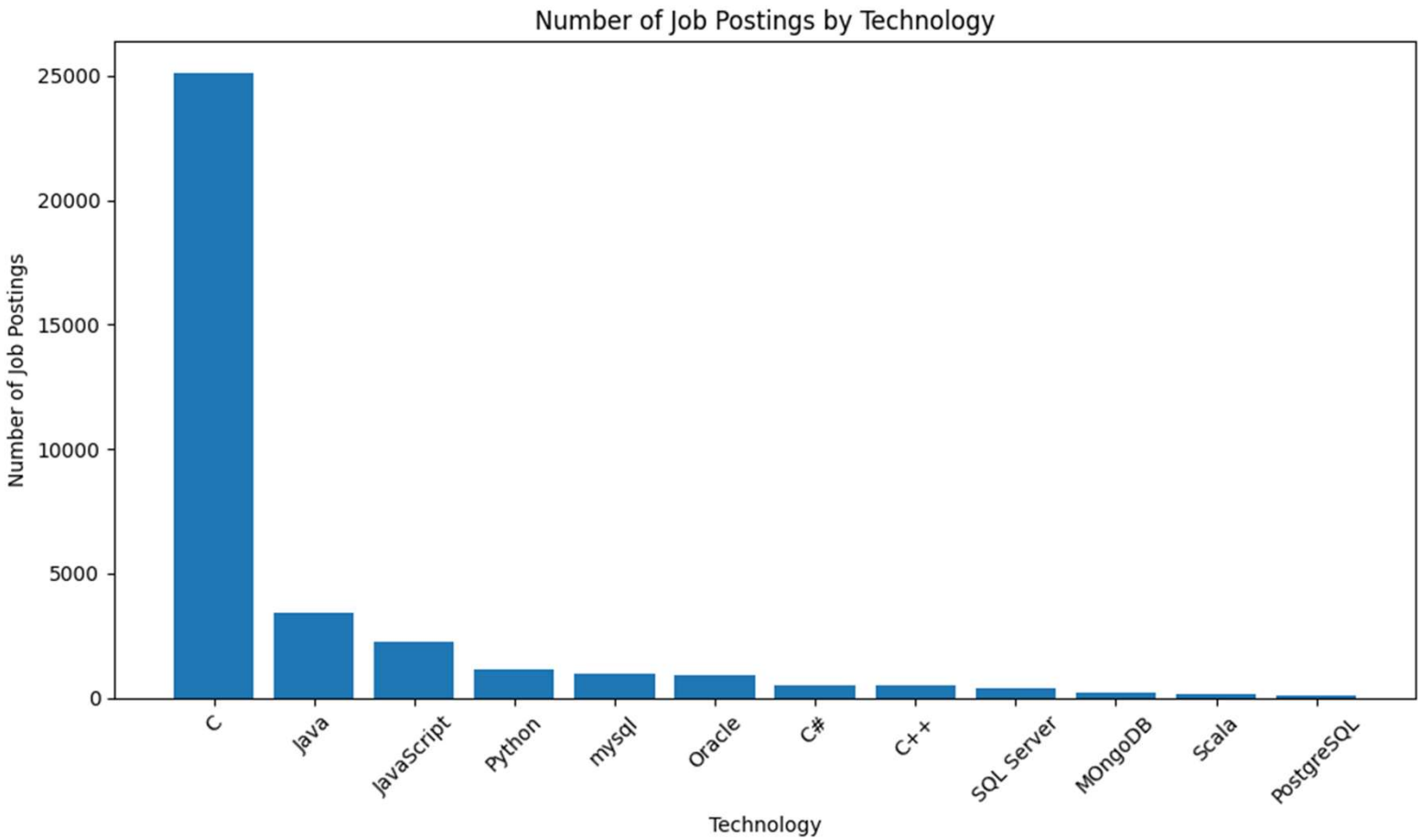
- Data for this project was collected from the **Stack Overflow Developer Survey dataset**.
- Additional datasets were gathered using **Job APIs and web scraping** to analyze job postings and programming language salaries.
- Data cleaning and preprocessing were performed using **Python, Pandas, and Jupyter Notebook**.
- Data wrangling steps included:
  - Handling missing values and cleaning multi-select survey columns.
  - Transforming and preparing data for visualization and dashboard creation using **IBM Cognos Analytics**.

Results

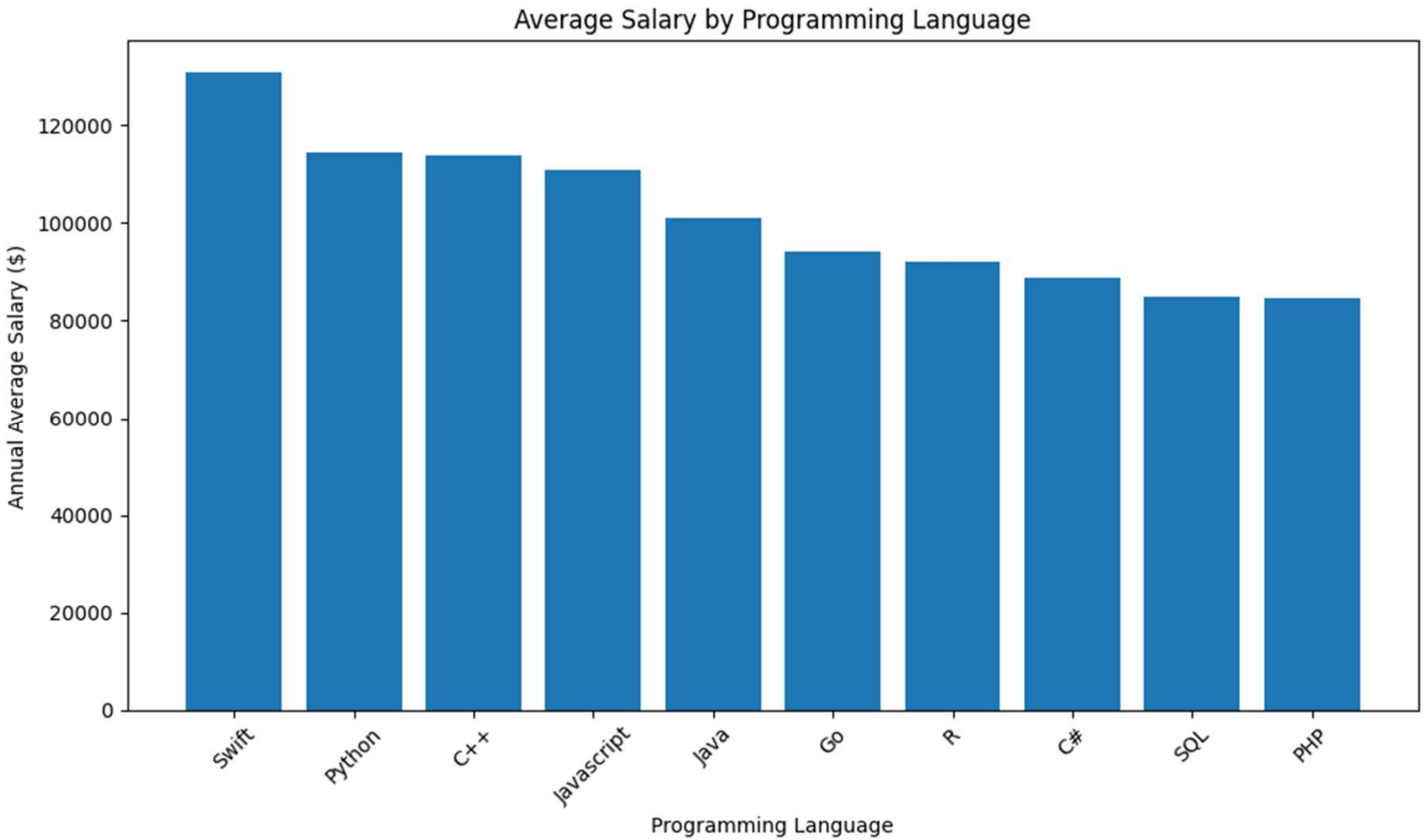


<Please present your results in the following slides.>

JOB POSTINGS



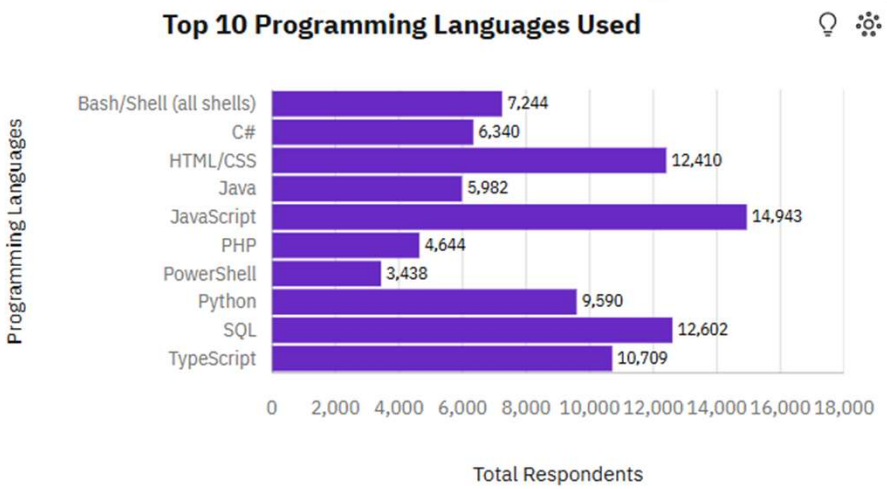
POPULAR LANGUAGES



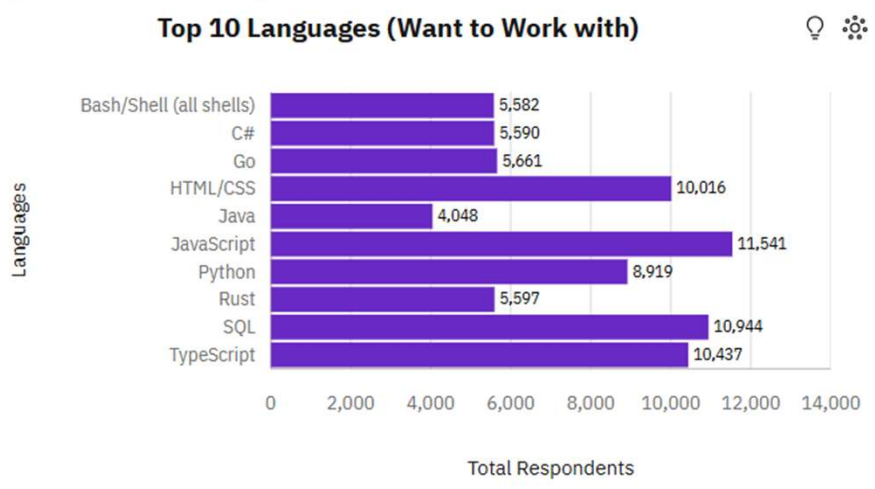
PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript, HTML/CSS, SQL, and Python are among the most widely used programming languages by developers today.
- Developers show strong interest in modern languages such as **Rust and TypeScript** for future projects.
- Python continues to grow in popularity in **data science, machine learning, and backend development**.

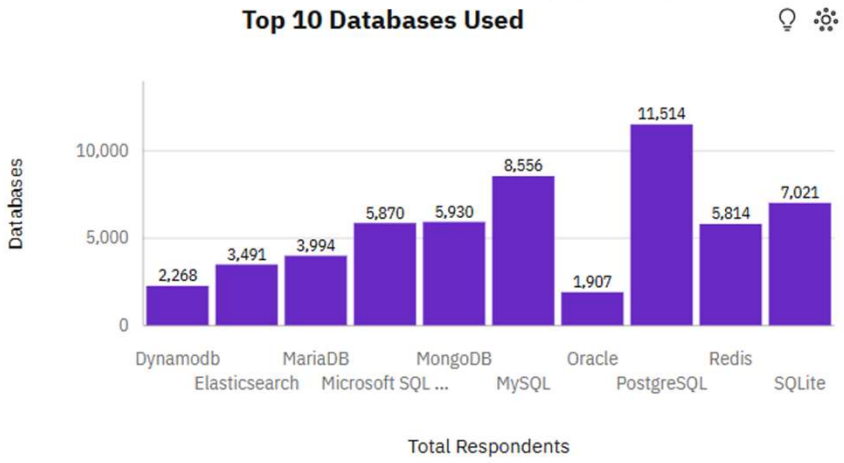
Implications

- Developers should focus on **modern and versatile programming languages** such as Python and JavaScript.
- Emerging languages like **Rust and TypeScript** may become increasingly important in the future.
- Continuous learning of new technologies is essential to stay competitive in the **rapidly evolving software industry**.

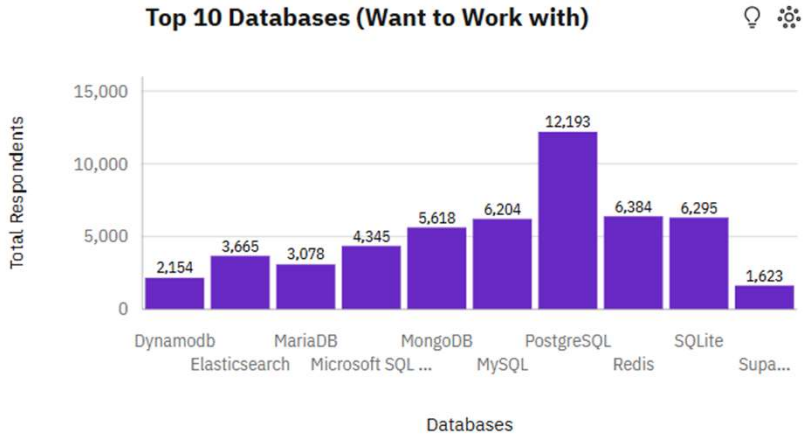
DATABASE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



- PostgreSQL, MySQL, and SQLite are among the most widely used databases by developers today.
- PostgreSQL shows the strongest interest among developers for future projects.
- Modern database technologies such as **Redis and MongoDB** continue to be widely adopted.

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

• PostgreSQL is one of the most popular databases both currently and for future use.
• MySQL and SQLite remain widely used due to their reliability and simplicity.
• NoSQL databases such as MongoDB and Redis continue to gain traction..

Implications

- Developers should gain strong knowledge of **PostgreSQL and MySQL**, as they remain highly relevant.
- Understanding **both relational and NoSQL databases** is important for modern applications.
- Organizations are increasingly adopting scalable database solutions for large data systems.

DASHBOARD

<Please present your dashboard in the following slides.>

DASHBOARD TAB 1: Current Technology Usage

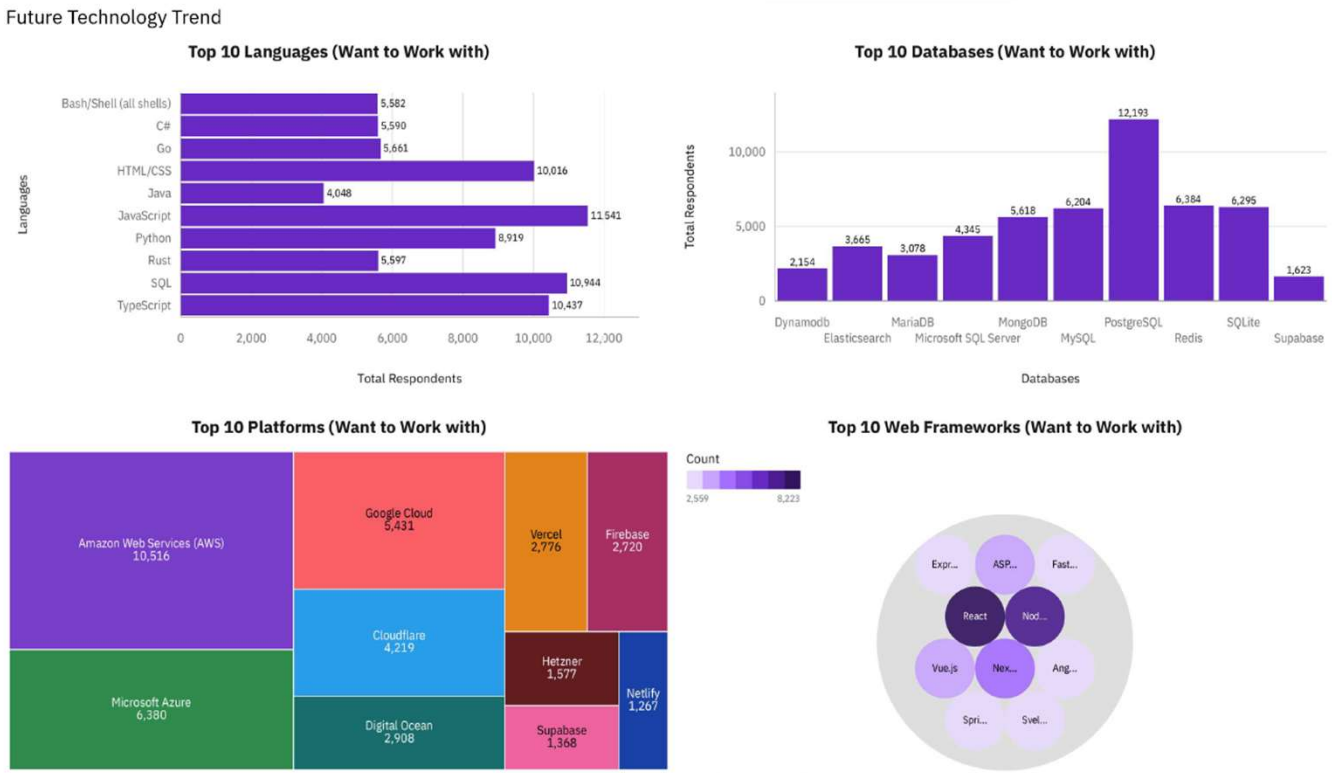
- JavaScript, SQL, HTML/CSS, and Python are among the most widely used programming languages by developers.

- PostgreSQL, MySQL, and SQLite are the most commonly used database technologies.

- Cloud platforms such as **Amazon Web Services (AWS)** and **Microsoft Azure** dominate platform usage among developers.

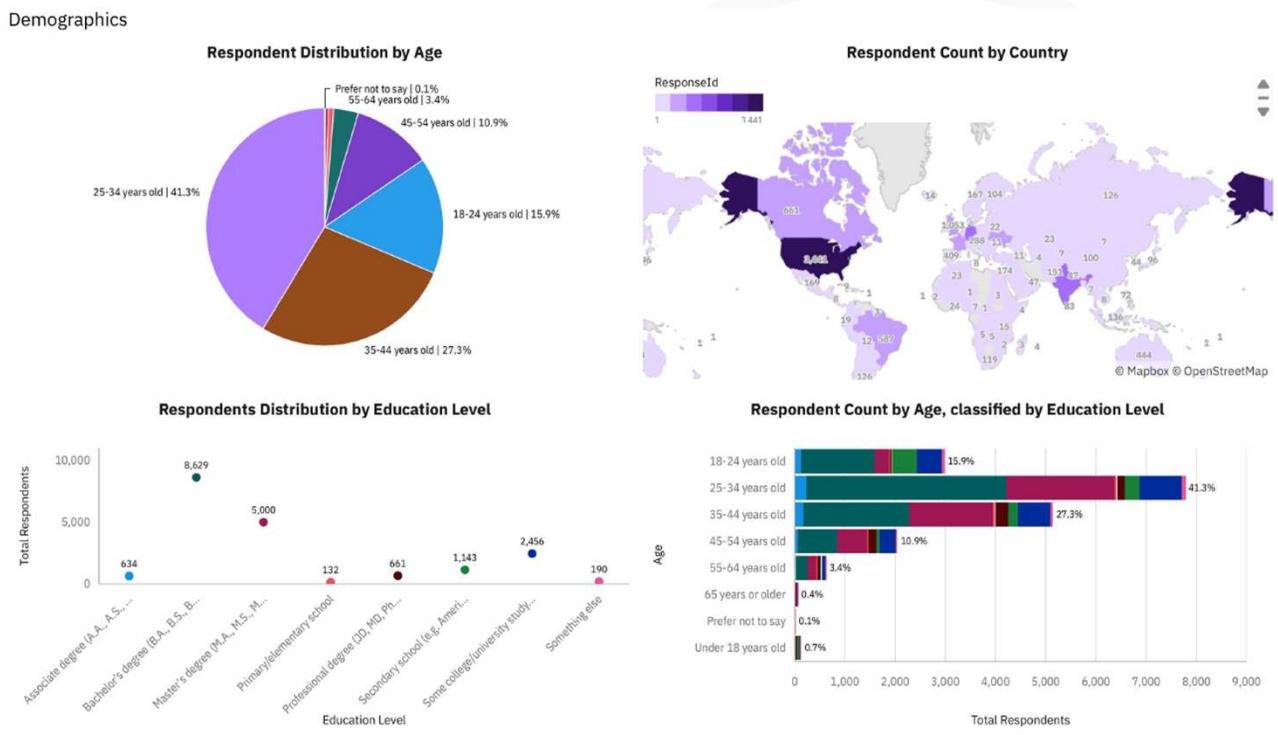
- Popular web frameworks include **React, Node.js, Angular, and Vue.js**, highlighting strong demand for modern web development technologies.

DASHBOARD TAB 2: Future Technology Trends



- Developers show strong interest in languages such as **JavaScript, SQL, Python, and TypeScript** for future work.
- **PostgreSQL** is the most desired database technology among developers, indicating growing interest in modern relational databases.
- Cloud platforms such as **Amazon Web Services (AWS)** and **Microsoft Azure** remain highly preferred for future development.
- Web frameworks such as **React, Node.js, and Next.js** continue to gain popularity for modern web application development.

DASHBOARD TAB 3: Demographics



- Most respondents are in the **25–34 age group**, followed by **35–44**, indicating a predominantly early-career developer population.

- The majority of developers hold a **Bachelor's or Master's degree**, showing that formal higher education is common in the tech workforce.

- The survey responses are **globally distributed**, with the **United States and several European and Asian countries** contributing the largest number of respondents.

DISCUSSION

- **JavaScript, SQL, Python, and HTML/CSS** remain the most widely used programming languages among developers today.
- Developers show strong interest in **JavaScript, TypeScript, Python, and SQL** for future work, indicating continued demand for these technologies.
- **PostgreSQL, MySQL, and SQLite** are among the most widely used databases, with **PostgreSQL gaining strong future interest**.
- Cloud platforms such as **AWS and Microsoft Azure** dominate platform usage, while frameworks like **React and Node.js** remain highly popular for modern web development.

OVERALL FINDINGS & IMPLICATIONS

Findings

- **JavaScript, SQL, and Python** are among the most widely used programming languages by developers.
- **PostgreSQL, MySQL, and SQLite** are the most commonly used database technologies.
- Most respondents are **25–34 years old and hold a Bachelor's or Master's degree**, representing an early-career developer population.

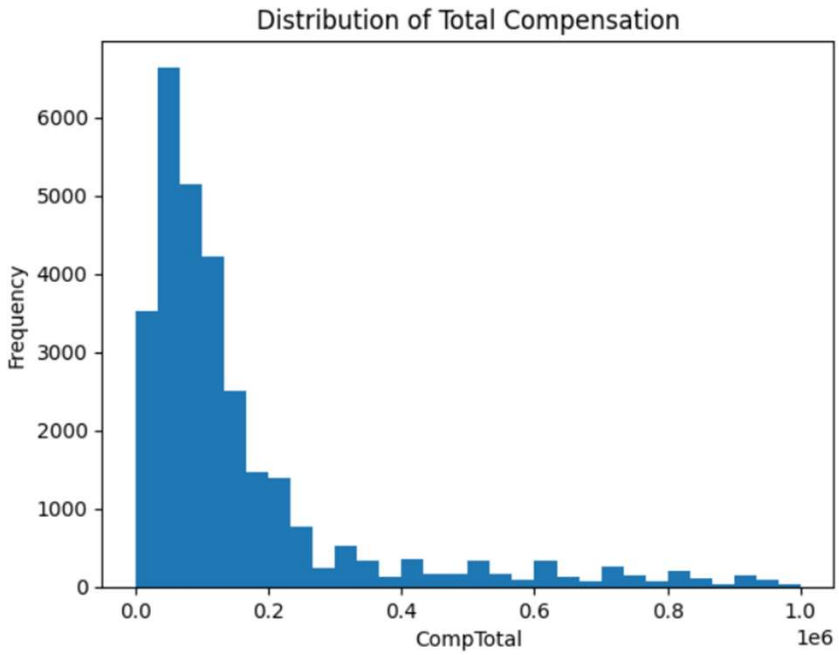
Implications

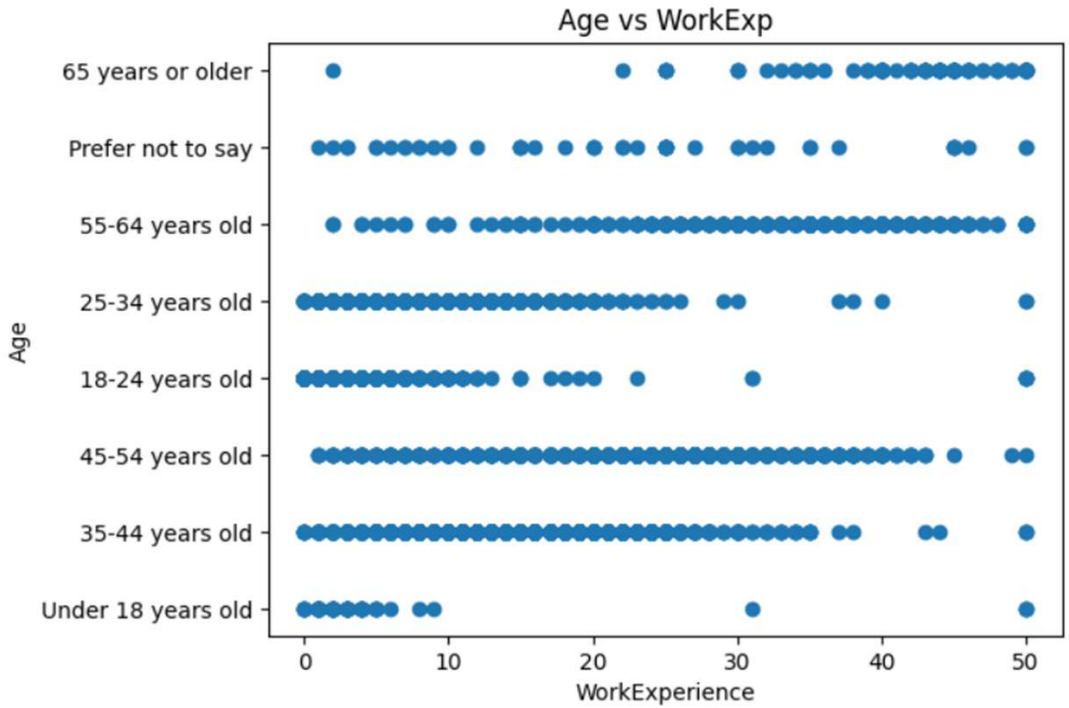
- Professionals should focus on **learning widely used languages such as JavaScript, Python, and SQL** to improve career opportunities.
- Organizations should invest in **modern databases like PostgreSQL and scalable cloud platforms such as AWS and Azure**.
- Training programs and universities should continue emphasizing **modern web technologies and cloud computing skills** to meet industry demand.

CONCLUSION

- • Modern development is strongly driven by **JavaScript, Python, SQL, and web technologies**.
- • **PostgreSQL and cloud platforms like AWS and Azure** are becoming increasingly important in the technology ecosystem.
- • Developers continue to show strong interest in **modern frameworks such as React and Node.js** for building scalable applications.
- • Understanding current technology usage and future trends helps **developers, organizations, and educators make better technology and career decisions**.

APPENDIX


Distribution of Total Compensation
Frequency
6000
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
1e6
CompTotal


Age vs WorkExp
Age
65 years or older
Prefer not to say
55-64 years old
25-34 years old
18-24 years old
45-54 years old
35-44 years old
Under 18 years old
0
10
20
30
40
50
WorkExperience